

Complete Hardware Summary – Humanoid Teleoperation Arm

Category	Component	Qty	Notes
Shoulder Actuators	XH540-V270-R	3	24 V Dynamixel
Elbow Actuators	MX-64R	2	12 V Dynamixel
Wrist/Hand Root	MX-28R	2	12 V Dynamixel
Motor Mounts	HN13-I101	1	For XH540
	HN07-I101	2	For MX-28
	HN05-I101	1	For MX-64
Palm Motors	Micro Metal Gearmotor HPCB 12V	6	With encoders
Palm Motor Drivers	TB6612FNG Dual Driver	3	Palm DC motor control
IMUs	BNO086	3	Upper arm, forearm, palm
Additional IMU	BNO085	1	Optional
Force Sensors	Flexible FSR 5 cm	6	Palm/finger sensing
Main MCU	Teensy 4.1	1	Primary control unit
Palm/Support MCU	Arduino Nano ESP32	3	Palm + auxiliary control
RS-485 Interfaces	U2D2 + Power Hub	3	Dynamixel communication
Bearings	QJ210	3	Main rotary joints
Main PSU	LRS-600-24	1	24 V 600 W central PSU
12 V Rail A	24→12 V Buck, 15A	1	MX64/MX28
12 V Rail B	24→12 V Buck, 15A	1	Palm motors
Logic Power	12→5 V Buck	1	5 V logic supply
Logic Power	S9V11F3S5 (12→3.3 V)	1	3.3 V rail
Headers	Arduino Nano Stackable Header Kit	3	For Nano ESP32
	1×40 Male Header 2.54mm	3	General pin headers
	JST-SH 4-pin Connector + wires	3	Sensor/IMU wiring
	Flexible Qwiic Cable 100mm	3	I2C sensor interconnect
	Teensy Stackable Header Kit (Extended)	1	For Teensy 4.1

This document lists all hardware components included in the final design.